Given the magnetic field intensity

$$\vec{H} = 2(y - z)\,\vec{a}_x + x^2\,\vec{a}_y - 3\,\ln\left(\frac{x}{y}\right)\vec{a}_z \quad A/m$$

Calculate the total current that passes through a square that lies in the positive x-y plane with one corner at the origin. The length of each side is 1 m.

Given the magnetic field intensity

$$\vec{H} = 2(y - z)\,\vec{a}_x + x^2\,\vec{a}_y - 3\,\ln\left(\frac{x}{y}\right)\vec{a}_z \quad A/m$$

Calculate the total magnetic force exerted on a piece of wire extending between (0, 0, 0) and (0, 2, 0) in free space and carrying a current of 20 A. All dimensions are in meters.

If allowed, in which direction will the wire move?

Consider a thick conducting rod of radius a and infinite length. The rod carries a constant current density of magnitude 0.1 A/m^2.

Use Ampere's law to find the magnetic field intensity at any point inside or outside the rod and sketch it.

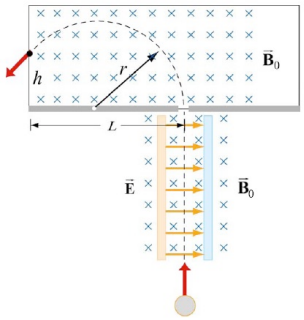
Given $\mathbf{H}$=10 $\mathbf{a}_\phi$ (A/m), find the total current passing through a circular contour of radius 10 cm placed in the xy plane and centered at the origin.

A very thin wire of length 0.2 m is brought in the neighborhood of a magnetic field given by

$$\vec{B} = 1.5x\,\vec{a}_x \qquad \text{Wb/m}^2$$

The wire is placed parallel to the y axis at x=2 m. If the wire carries a constant current I, what is the force on the wire?

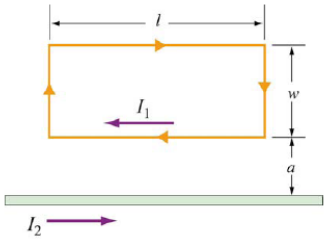
A charged particle of charge q, mass m and speed v enters from the bottom of the spectrometer as shown in the figure. The magnetic field is everywhere B_o, but the electric field E is only in the region where the trajectory of the particle is a straight line (lower part). Both fields are constant. (Ignore gravity).



What is the sign of the particle charge?
What should be the value of the electric field?
Show that the radius r of the particle trajectory in the upper region is given by

$$r = \frac{mv}{qB_o}$$

The wire loop shown below carries a constant current I_1 and is located near an infinite wire that carries constant current I_2. Indicate the direction of the net force on the wire loop due to the presence of the infinite wire. Explain.



A very thin wire of length 0.2 m is brought in the space of a magnetic field given by

$$\vec{B} = 1.5x\,(\vec{a}_x + \vec{a}_y) \quad \text{Wb/m}^2$$

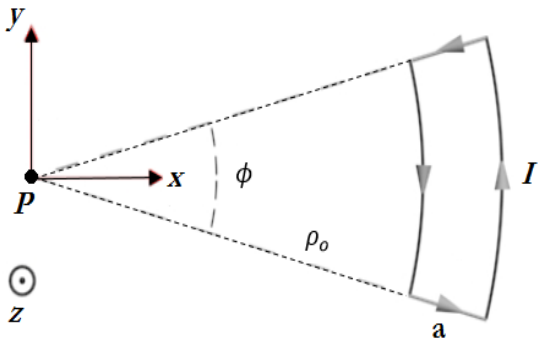
The wire is placed parallel to the *y*-axis at $x = 2$ m. If the wire carries a constant current $I = 0.1$ A, what is the magnitude and direction of the force on the wire?

The current density in a certain cylindrical space is given by

$$\vec{J} = \frac{100}{\rho^2}\,\vec{a}_\rho \ (\text{A/m}^2)$$

Calculate the rate of change in the volume charge density.
Does the rate in (a) indicate an increase of a decrease of charge density with time?
Calculate the total current passing through the surface defined by $-1 < z < 1$ and $\rho = 2$.

For the structure shown in the figure, the current I is constant and flows in the indicated direction. The radius of the inner arc segment is ρ_o and that of the outer arc segment is $\rho_o + a$. Point P is the common center for both arc segments.



Using Biot-Savart Law, find the magnetic field at point *P* due to the inner arc segment.
Find the total magnetic field at point *P*.
Sketch the magnetic field magnitude versus a, as the value of a changes from 0 to $a \gg \rho_o$. Explain the result.

The excess free charge density in a certain homogeneous medium decreases to one-third of its initial value in 20 μs. The conductivity of the medium is 10^{-5} S/m. What is the dielectric constant of the medium.
